

ECE 3210 hw02 [SOLUTIONS]

1. Consider the system $y(t) = x(2t)$ with input $x(t) = |t|$. Let $T = 1$ and define

$$\tilde{y}(t) = H\{x(t - T)\}.$$

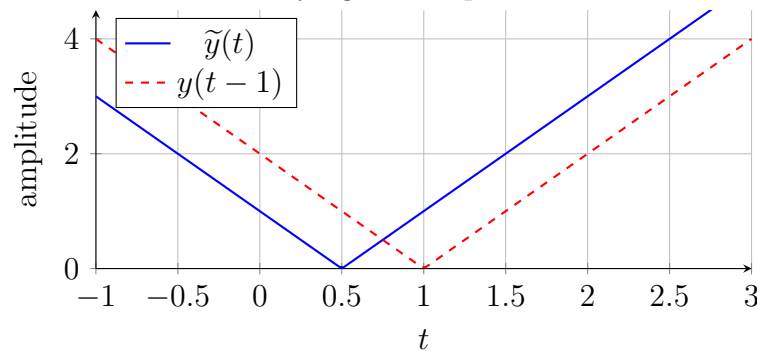
- (a) Find $y(t)$, $\tilde{y}(t)$, and $y(t - T)$. Plot each of the three signals on the same axes.

Solution:

$$y(t) = x(2t) = 2|t|, \quad \tilde{y}(t) = x(2t - 1) = |2t - 1|, \quad y(t - T) = 2|t - 1|.$$

- (b) Is the system time invariant? Hint: compare $\tilde{y}(t)$ and $y(t - T)$. If they are not equal, then the system is not time invariant.

Solution: Not time-invariant. Since $|2t - 1| \neq 2|t - 1|$ in general, delaying the input is not the same as delaying the output.



2. Consider the real LTIC system

$$(D^2 + 5D + 6)y(t) = (D^2 + 7D + 11)x(t).$$

(a) What is the order n of the system?

Solution: The highest derivative of $y(t)$ is second order, so $n = 2$.

(b) Find the characteristic equation $Q(\lambda) = 0$.

Solution: Replacing D in the left-side operator by λ gives $Q(\lambda) = \lambda^2 + 5\lambda + 6 = 0$. The roots are $\lambda = -2$ and $\lambda = -3$.

(c) Find the quasi-zero-input response $y_n(t)$.

Solution: The natural modes are e^{-2t} and e^{-3t} . With the specified normalization,

$$y_n(t) = e^{-2t} - e^{-3t}.$$

(d) Find the b_n term.

Solution: The coefficient of $D^2x(t)$ is $b_n = 1$.

(e) Find the impulse response $h(t)$.

Solution: We use the equation $h(t) = b_n\delta(t) + P(D)y_n(t)u(t)$, where $P(D)$ is the right-side operator of the system. In this case, $P(D) = D^2 + 7D + 11$. We should first focus on finding $P(D)y_n(t)$. We have

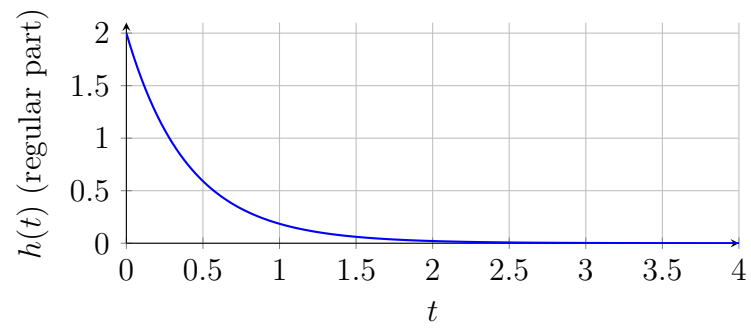
$$y_n(t) = e^{-2t} - e^{-3t}, \quad y'_n(t) = -2e^{-2t} + 3e^{-3t}, \quad y''_n(t) = 4e^{-2t} - 9e^{-3t}.$$

We can now compute $P(D)y_n(t)$

$$\begin{aligned} P(D)y_n(t) &= y''_n(t) + 7y'_n(t) + 11y_n(t) \\ &= (4e^{-2t} - 9e^{-3t}) + 7(-2e^{-2t} + 3e^{-3t}) + 11(e^{-2t} - e^{-3t}) \\ &= 1e^{-2t} + 1e^{-3t} \\ &= e^{-2t} + e^{-3t}. \end{aligned}$$

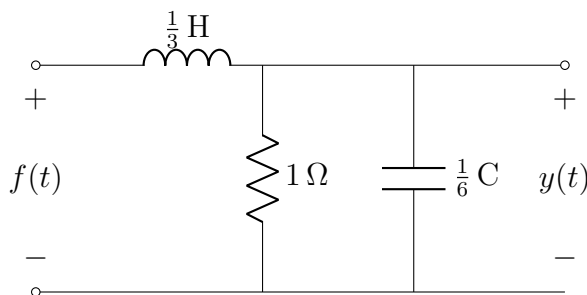
Putting it all together gives

$$h(t) = \delta(t) + (e^{-2t} + e^{-3t})u(t).$$



The arrow at $t = 0$ represents the unit impulse $\delta(t)$.

3. For the circuit below, with input $f(t)$ and output $y(t)$, answer the following questions.



- (a) Determine the ODE governing the system.

Solution: The easiest approach to the circuit analysis is a basic KCL from the node at the top

$$i_L + i_R + i_C = 0 \text{ A.}$$

We know that $i_L = 3 \int_{-\infty}^t (y(\tau) - f(\tau)) d\tau$, $i_R = y(t) - f(t)$, and $i_C = \frac{1}{6} \frac{dy}{dt}$. Substituting these into the KCL equation gives

$$3 \int_{-\infty}^t (y(\tau) - f(\tau)) d\tau + y(t) + \frac{1}{6} \frac{dy}{dt} = 0.$$

Taking the derivative of both sides gives

$$3(y(t) - f(t)) + y'(t) + \frac{1}{6} y''(t) = 0,$$

which can be rearranged to give

$$y''(t) + 6y'(t) + 18y(t) = 18f(t).$$

- (b) Determine the zero-input response $y_n(t)$.

Solution: We see that the characteristic equation $\lambda^2 + 6\lambda + 18 = 0$ has roots $-3 \pm 3j$. Hence, the zero-input response is

$$y_n(t) = e^{-3t}(A \cos(3t) + B \sin(3t)).$$

Using the initial conditions $y(0) = 0$ and $y'(0) = 1$, we can solve for A and B . We have

$$y_n(0) = A = 0,$$

and so $y_n(t) = B e^{-3t} \sin(3t)$. If we take the derivative, we have

$$y'_n(t) = B e^{-3t} (3 \cos(3t) - 3 \sin(3t)),$$

and applying the initial condition

$$y'_n(0) = 3B = 1 \implies B = \frac{1}{3}.$$

Using this we have

$$y_n(t) = \frac{1}{3}e^{-3t} \sin(3t).$$

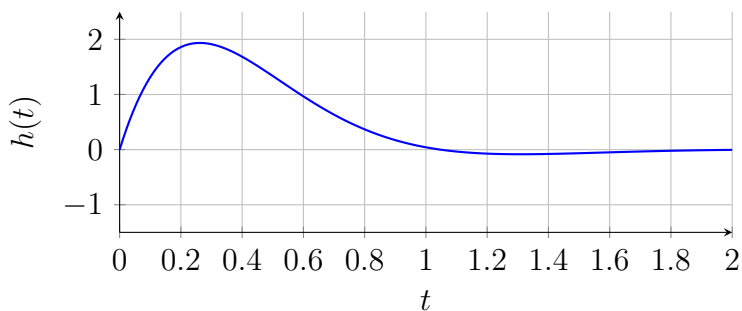
(c) Determine the impulse response $h(t)$.

Solution: Using the equation $h(t) = b_n\delta(t) + P(D)y_n(t)u(t)$, where $P(D)$ is the right-side operator of the system, we have $P(D) = 18$. Hence,

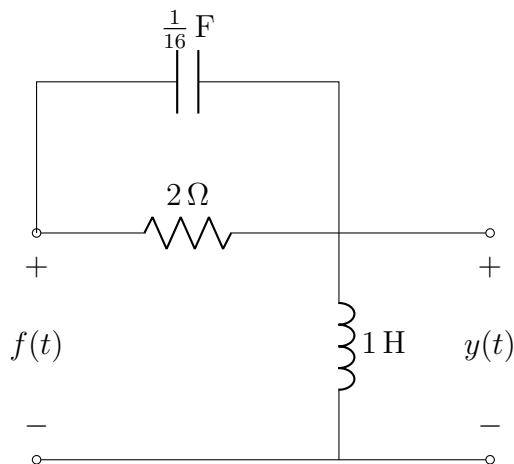
$$h(t) = 18 \cdot \frac{1}{3}e^{-3t} \sin(3t)u(t)$$

and so

$$h(t) = 6e^{-3t} \sin(3t)u(t).$$



4. For the circuit below, with input $f(t)$ and output $y(t)$, answer the following questions.



- (a) Determine the ODE governing the system.

Solution: Using KCL at the top node, we have

$$i_R + i_C + i_L = 0.$$

We know that $i_R = \frac{y(t)-f(t)}{2}$, $i_C = \frac{1}{16} \frac{d}{dt}(y(t) - f(t))$, and $i_L = \int_{-\infty}^t y(\tau) d\tau$. Substituting these into the KCL equation gives

$$\frac{1}{2}(y(t) - f(t)) + \frac{1}{16} \frac{d}{dt}(y(t) - f(t)) + \int_{-\infty}^t y(\tau) d\tau = 0.$$

Taking the derivative of both sides gives

$$\frac{1}{2}(y'(t) - f'(t)) + \frac{1}{16}(y''(t) - f''(t)) + y(t) = 0,$$

which can be rearranged to give

$$y''(t) + 8y'(t) + 16y(t) = f''(t) + 8f'(t).$$

- (b) Determine the zero-input response $y_n(t)$ (as if we were trying to find $h(t)$).

Solution: The characteristic equation is $\lambda^2 + 8\lambda + 16 = 0$. The repeated root is -4 , yielding

$$y_n(t) = e^{-4t}(A + Bt).$$

Applying the initial conditions $y_n(0) = 0$ and $y'_n(0) = 1$, we get $A = 0$ and $B = 1$, so

$$y_n(t) = te^{-4t}.$$

(c) Determine the impulse response $h(t)$.

Solution: Using the equation $h(t) = b_n\delta(t) + P(D)y_n(t)u(t)$, where $P(D)$ is the right-side operator of the system, we have $P(D) = D^2 + 8D$. Hence,

$$h(t) = \delta(t) + (D^2 + 8D)te^{-4t}u(t).$$

We need to solve the derivatives of $te^{-4t}u(t)$. Using the product rule, we have

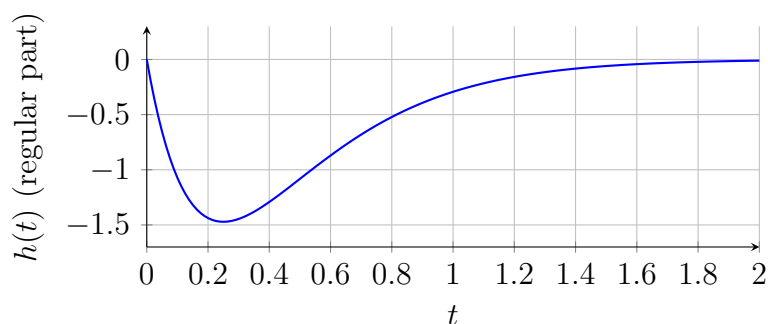
$$\frac{d}{dt}te^{-4t} = e^{-4t} - 4te^{-4t}.$$

Using the product rule again, we have

$$\frac{d^2}{dt^2}te^{-4t} = -4e^{-4t} - 4(e^{-4t} - 4te^{-4t}) = -8e^{-4t} + 16te^{-4t}.$$

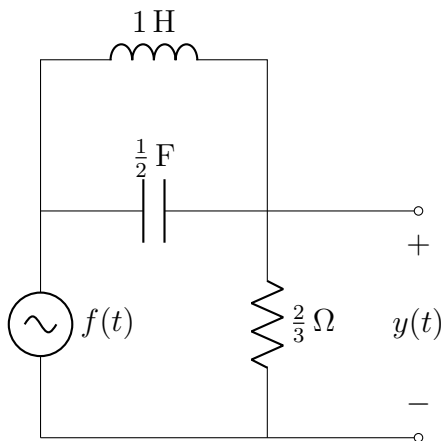
Hence, we have

$$\begin{aligned} h(t) &= \delta(t) + (y_n''(t) + 8y_n'(t))u(t) \\ &= \delta(t) + (-8e^{-4t} + 16te^{-4t} + 8(e^{-4t} - 4te^{-4t}))u(t) \\ &= \delta(t) - 16te^{-4t}u(t). \end{aligned}$$



The arrow at $t = 0$ represents $\delta(t)$.

5. For the circuit below, derive the input-output relation and answer the following questions.



- (a) Determine the ODE governing the input-output relation of the circuit.

Solution: We can start with a KCL at the node between the inductor, capacitor, and resistor. The current through the inductor is $i_L = \int_{-\infty}^t (y(\tau) - f(\tau)) d\tau$, the current through the capacitor is $i_C = C \frac{d}{dt}(y(t) - f(t))$, and the current through the resistor is $i_R = \frac{y(t)}{R}$. Applying KCL gives

$$i_L + i_C + i_R = 0 \text{ A.}$$

Applying the derivative terms gives

$$\int_{-\infty}^t (y(\tau) - f(\tau)) d\tau + \frac{1}{2}(y'(t) - f'(t)) + \frac{3}{2}y(t) = 0.$$

Taking the derivative of both sides gives

$$y(t) - f(t) + \frac{1}{2}(y''(t) - f''(t)) + \frac{3}{2}y'(t) = 0,$$

Rearranging gives

$$y'' + 3y' + 2y = f'' + 2f$$

- (b) Given $y(0^-) = 0$ and $y'(0^-) = 1$, determine the zero-input response $y_n(t)$.

Solution: We see the characteristic equation is $\lambda^2 + 3\lambda + 2 = 0$, which has roots $\lambda_1 = -1$ and $\lambda_2 = -2$. Hence the natural response has the form $Ae^{-t} + Be^{-2t}$. Applying the initial conditions

$$y_n(0) = A + B = 0$$

and

$$y'_n(0) = -A - 2B = 1.$$

Solving this system of equations gives $A = 1$ and $B = -1$, hence $y_n(t) = e^{-t} - e^{-2t}$.

(c) Determine the impulse response $h(t)$.

Solution: We apply the equation $h(t) = b_n\delta(t) + P(D)y_n(t)u(t)$ where $P(D) = D^2 + 2$. Applying this operator to the natural response gives

$$P(D)y_n(t) = y''_n(t) + 2y_n(t).$$

Finding

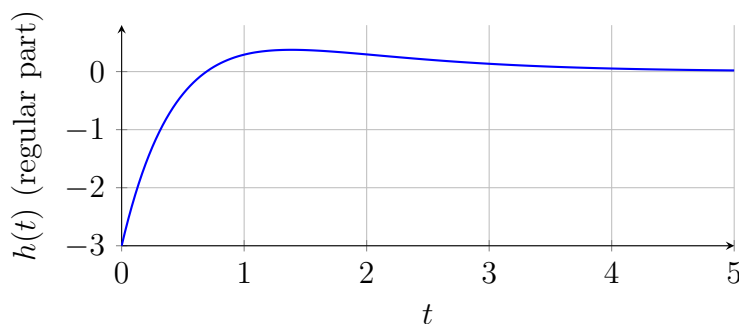
$$y'_n(t) = -e^{-t} + 2e^{-2t}.$$

and

$$y''_n(t) = e^{-t} - 4e^{-2t}.$$

Hence

$$\begin{aligned} h(t) &= \delta(t) + (y''_n(t) + 2y_n(t))u(t) \\ &= \delta(t) + (e^{-t} - 4e^{-2t} + 2(e^{-t} - e^{-2t}))u(t) \\ &= \delta(t) + (3e^{-t} - 6e^{-2t})u(t). \end{aligned}$$



The arrow at $t = 0$ represents $\delta(t)$.